Optics Final Project

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Introduction

In class, we developed our understanding of several Optical frameworks that work very well for problems with discontinuous changes in refractive index. However, many optical phenomena require an understanding of how light behaves in a material with a continuously varying index of refraction. The most common way to understand these situations uses a Hamiltonian formulation of Optics, with many possible layers of abstraction.

This report lays out the fundamentals of the approach and motivates higher level concepts that simplify non-exact calculations (Poisson brackets and Lie transformations). By analyzing the problem of a gradient index material in two ways, we build intuition for what these higher level mathematical tools allow us to do.

1 Hamiltonian Optics

Any optical situation (that doesn't involve rays perpendicular to the optical axis) can be parameterized in a specific way that lends itself nicely to analysis using Hamiltonian Mechanics [1]. To fully describe the three-dimensional space, we use x_i, the usual cartesian coordinates. By convention, x_1 and x_2 represent the spatial coordinates, while x_3 represents the optical axis, but we've seen in class that propagation along this axis is on unequal footing with the spatial coordinates. Namely, evolving along this axis is more similar to passing through time than space, so we define $\tau = x_3$.

We then define the optical Lagrangian to be

$$L = n(x_1, x_2, \tau)\sqrt{1 + \dot{x}_1 + \dot{x}_2} \tag{1}$$

where $\dot{Z} = \frac{\partial Z}{\partial \tau}$. The Euler-Lagrange equations take the same form as in classical mechanics and allow one to analyze optical problems. However, the more common approach in optics is to transition to the Hamiltonian formalism.

We start by defining optical momenta

$$p_i = \frac{\partial L}{\partial \dot{x}_i},$$

which allow us find the optical Hamiltonian from the Lagrangian (see [1] for a full derivation). We now change from Cartesian coordinates, x_i, to generalized coordinates, q_i, where i is either 1 or 2.

Then, we define the Hamiltonian of a general optical system to be

$$H = -\sqrt{n(\mathbf{q})^2 - \mathbf{p}^2}. \tag{2}$$

Hamilton's equations take the same form as in classical mechanics

$$\frac{\partial H}{\partial q_i} = -\dot{p}_i \tag{3}$$

$$\frac{\partial H}{\partial p_i} = \dot{q}_i. \tag{4}$$

With the basics of this framework in place, we can now solve an optical problem.

1.1 Example: GRIN

Suppose we wish to characterize the motion of light through an optical fiber with a continuously varying index of refraction

$$n(\mathbf{q}) = \sqrt{n_0^2(1 - \mathbf{q}^2) + \mathbf{q}^2},$$

where n_0 is greater than 1 [2]. The Hamiltonian of this system is then

$$H = -\sqrt{n_0^2(1 - \mathbf{q}^2) + \mathbf{q}^2 - \mathbf{p}^2},$$

which gives the governing system of equations:

$$\begin{aligned} -\dot{\mathbf{p}} &= \frac{\partial H}{\partial \mathbf{q}} = \frac{-\mathbf{q}(1 - n_0^2)}{\sqrt{-n_0^2\left(\mathbf{q}^2 - 1\right) - \mathbf{p}^2 + \mathbf{q}^2}} = \frac{\mathbf{q}(1 - n_0^2)}{H} \\ \dot{\mathbf{q}} &= \frac{\partial H}{\partial \mathbf{p}} = \frac{\mathbf{p}}{\sqrt{-n_0^2\left(\mathbf{q}^2 - 1\right) - \mathbf{p}^2 + \mathbf{q}^2}} = \frac{-\mathbf{p}}{H}. \end{aligned}$$

We can now solve this system, leveraging the time-independence of the Hamiltonian to give the following expressions for motion, given an initial position of $\mathbf{q}_0$ and momentum of $\mathbf{p}_0$:

$$\mathbf{q}(\tau) = \mathbf{q}_0 \cos \omega\tau - \frac{\mathbf{p}_0}{H\omega} \sin \omega\tau \tag{5}$$

$$\mathbf{p}(\tau) = \mathbf{p}_0 \cos \omega\tau + \mathbf{q}_0 H\omega \sin \omega\tau \tag{6}$$

where ω is defined to be $\sqrt{n_0^2 - 1}/H$.

Choosing initial conditions $\mathbf{q}_0 = 0.8$, $\mathbf{p}_0 = 0$, we plot the position of a particle and its optical momentum as a function of position along an optical fiber (τ). Since the fiber is cylindrically symmetric, we plot only in two dimensions.

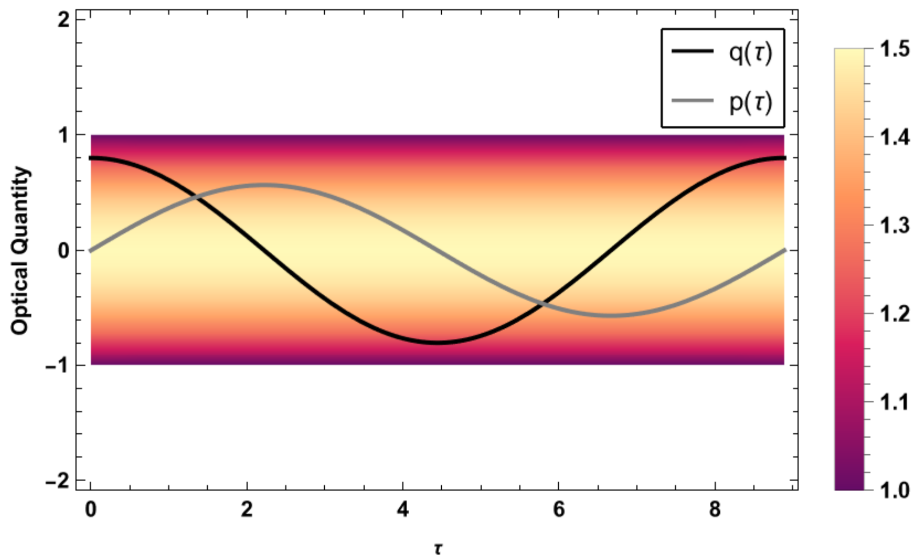

Figure 1: Evolution of **p** and **q** as light travels along an optical fiber of varying refractive index (fiber extends from -1 to 1). The index of the fiber at a point is given by the color bar at right.

1.2 Liouville's Theorem

An important result from Hamiltonian Mechanics is Liouville's theorem, which says that the area enclosed by several points, corresponding to the motion of a system of particles, is constant during that motion [3]. We shall examine the motion of points in the geometry above and show that Liouville's theorem holds.

Consider four points in phase space that comprise a square at $\tau = 0$, with initial conditions explicitly laid out in Table 1:

Point	$q(0)$	$p(0)$
A	q_0	p_0
B	$q_0 + \epsilon$	p_0
C	$q_0 + \epsilon$	$p_0 + \delta$
D	q_0	$p_0 + \delta$

Table 1: Initial Conditions in phase space

We now consider their evolution through phase space over time. At a given time, they will form a parallelogram, as in Figure 2.

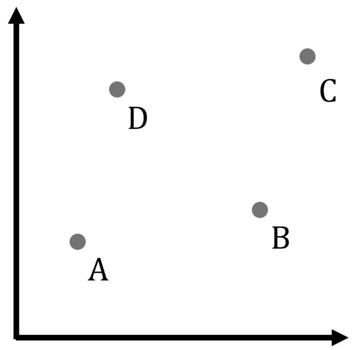

Figure 2: Four points in phase space at a given time

We wish to calculate the area of the parallelogram formed by these points at any arbitrary time. The simplest means of calculating this area is finding the magnitude of the cross product: the area equals $|\overrightarrow{AB}\times\overrightarrow{AD}|$. Given the initial conditions above, $\overrightarrow{AB} = (\epsilon\cos\omega\tau, H\omega\epsilon\sin\omega\tau)$, while $\overrightarrow{AD} = (-\frac{\delta}{H\omega}\sin\omega\tau, \delta\cos\omega\tau)$. Thus, we find

$$\begin{aligned} A = |\overrightarrow{AB}\times\overrightarrow{AD}| &= (\epsilon\cos\omega\tau)(\delta\cos\omega\tau) - (H\omega\epsilon\sin\omega\tau)(-\frac{\delta}{H\omega}\sin\omega\tau) \\ &= \epsilon\delta(\cos^2\omega\tau + \sin^2\omega\tau) \\ &= \epsilon\delta \end{aligned}$$

so the area is constant in time, and Liouville's theorem holds for this optical case.

2 The Poisson Bracket

Analyzing Hamiltonian Optics with the above method reduces to solving slightly different forms of the same differential equations over and over again. But what if we could find some sort of invariant quantity that allowed us to simplify these calculations or solve them for a more general case? In Classical Mechanics, defining the Poisson Bracket clarifies which quantities are conserved over some path through our abstract space [4]. Analyzing the same structure here should yield a similar notion of conserved quantities, which we can use to model motion.

We define the Poisson Bracket of two functions f and g to be

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i}\right) \tag{7}$$

and say that two operators "Poisson commute" if the value of $\{f, g\} \equiv 0$. There are several important properties of Poisson Brackets:

$$\begin{aligned}
\{f, g\} &= -\{g, f\} \\
\{fg, h\} &= \{f, h\}g + f\{g, h\} \\
\{af + bg, h\} &= a\{f, h\} + b\{g, h\} \\
\{f, \{g, h\}\} + \{g, \{h, f\}\} &+ \{h, \{f, g\}\} \equiv 0.
\end{aligned}$$

Each of these follows directly from the differential nature of the Poisson bracket. The fourth identity is called the Jacobi identity and points to the more advanced mathematical framework of a Lie Algebra, which will not be fully explored here. For our purposes, it is taken as a useful fact.

A key result from classical mechanics is that the Poisson Bracket of a function on phase space (f) with the Hamiltonian is equal to its time derivative:

$$\{f, H\} = \frac{df}{d\tau}. \tag{8}$$

Thus, if the function f Poisson commutes with the Hamiltonian, it is a conserved quantity over time

$$\{f, H\} = \frac{df}{d\tau} = 0.$$

2.1 Finding an Invariant

We now seek a conserved quantity by analyzing the Poisson bracket of some function with the Hamiltonian. Let S be a generic function such that $S = S(q, p)$. Then, the Poisson bracket is

$$\begin{aligned}
\{S, H\} &= \frac{\partial S}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial S}{\partial p}\frac{\partial H}{\partial q} \\
&= S_q\left(\frac{p}{\sqrt{n(q,\tau)^2 - p^2}}\right) - S_p\left(\frac{\partial n}{\partial q}\frac{-n(q,\tau)}{\sqrt{n(q,\tau)^2 - p^2}}\right)
\end{aligned}$$

where $S_x = \frac{\partial S}{\partial x}$. We seek a function, S, such that the Poisson bracket vanishes, so

$$\begin{aligned}
S_q\left(\frac{p}{\sqrt{n(q,\tau)^2 - p^2}}\right) &= -S_p\left(\frac{\partial n}{\partial q}\frac{n(q,\tau)}{\sqrt{n(q,\tau)^2 - p^2}}\right) \\
pS_q &= -n(q,\tau)\frac{\partial n}{\partial q}S_p
\end{aligned}$$

which is a PDE of the function S. We can solve this PDE (see Equation 12 in Appendix A) for the following solution:

$$S(q, p) = S_0 \exp\left[\frac{p^2 - n^2}{2}\mu\right]$$

We note however that $n^2 - p^2$ is the negative square of the Hamiltonian of the system. Thus, our new invariant is

$$S(q,p) = S_0 \exp\left[\frac{-H^2}{2}\mu\right], \tag{9}$$

which we see is time independent. We now have a function that we can use in general for any problem characterized by a given optical Hamiltonian. In gravitational physics, we used invariants of a system to derive a "radial equation", which simplified the problem of motion on curved spacetime. Importantly, the problem went from solving a PDE to solving a system of ODEs, without any sort of discretization step. A similar approach seems possible here, but it would require a more complete understanding of the second invariant of the system.

Holm and Wolf [5] lay out a more comprehensive treatment of this idea, but limit their analysis to an invariant that exists for axisymmetric optical fibers. If one applied Noether's Theorem to the general optical Lagrangian (Equation 1), they should find more symmetries of the optical space, which yields another invariant along the optical path. I ran out of time to fully delve into this idea though.

3 Lie Operators

Another type of abstraction, unrelated to invariants, comes from introducing ideas from Lie theory. Consider a one-dimensional optical situation where we want to probe a point's evolution along the optical axis. Let us define the vector $X = (q, p)$, so that the entire problem can be characterized by

$$X'(\tau) = \{H, X(\tau)\}.$$

However, we note that this expression can be modified to yield a more suggestive form of the problem:

$$X'(\tau) = \{H, \cdot\}X(\tau)$$

where the second slot of the bracket acts on the first function right of the bracket. Then, this reads as a differential equation for $X(\tau)$ with some additional abstract operator. If we assume the Hamiltonian does not depend on τ, this is a constant operator and we know the solution to be

$$X(\tau) = e^{\tau\{H,\cdot\}}X_0. \tag{10}$$

This expression obscures the nuance that comes from exponentiating an operator (which can be treated in the same way as exponentiating a matrix). Thus,

$$e^{\tau\{H,\cdot\}} = \sum_n \frac{(-\tau)^n}{n!}\{H,\cdot\}^n$$

where the nth power of a Poisson bracket refers to one that has been nested n times. The implications of this approach are best understood through an example.

3.1 The GRIN revisited

We shall consider again a gradient index fiber optic, described by a Hamiltonian,

$$H = -\sqrt{n_0^2(1-q^2)+q^2-p^2}.$$

The positional evolution of this system can be understood by analyzing

$$q'(\tau) = \{H,\cdot\}q(\tau),$$

as laid out in [6]. The Lie transformation we seek is

$$\hat{Q} = e^{\{H,\cdot\}}q \equiv \sum_n \frac{(-\tau)^n}{n!}\{H,q\}^n$$

so we must evaluate the nested Poisson brackets. First, we lay out two important results

$$\begin{aligned}
\{H,q\} &= \frac{\partial H}{\partial q}\frac{\partial q}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial q}{\partial q} = \frac{p}{H}\\
\{H,p\} &= \frac{\partial H}{\partial q}\frac{\partial p}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial p}{\partial q} = -\frac{q(n_0^2-1)}{H},
\end{aligned}$$

which allow us to easily calculate the nested brackets:

$$\begin{aligned}
&\{H,q\}^0 = 1\\
&\{H,q\}^1 = \frac{p}{H}\\
&\{H,q\}^2 = \{H,\frac{p}{H}\} = -\frac{q(n_0^2-1)}{H^2}\\
&\{H,q\}^3 = \{H,-\frac{q(n_0^2-1)}{H^2}\} = -\frac{p(n_0^2-1)}{H^3}\\
&\{H,q\}^4 = \{H,-\frac{p(n_0^2-1)}{H^3}\} = \frac{q(n_0^2-1)^2}{H^4}.
\end{aligned}$$

Combining these elements of the Lie transformation with the governing equation of the system (Equation 10), we see

$$\begin{aligned}
q(\tau) &= 1 - \Big(\frac{p}{H}\tau\Big) + \Big(-\frac{q(n_0^2-1)}{H^2}\frac{\tau^2}{2}\Big) - \Big(-\frac{p(n_0^2-1)}{H^3}\frac{\tau^3}{6}\Big) + \Big(\frac{q(n_0^2-1)^2}{H^4}\frac{\tau^4}{24}\Big) + ...\\
&= 1 - \frac{q}{2}\Big((n_0^2-1)\Big(\frac{\tau}{H}\Big)^2\Big) + \frac{q}{24}\Big((n_0^2-1)^2\Big(\frac{\tau}{H}\Big)^4\Big) + ...\\
&- p\Big(\frac{\tau}{H}\Big) + \frac{p}{6}\Big((n_0^2-1)\Big(\frac{\tau}{H}\Big)^3\Big) + ...
\end{aligned}$$

which we recognize as the Taylor series of cosine and negative sine. Then,

$$\begin{aligned}
q(\tau) &= q\cos\Big(\frac{n_0^2-1}{H^2}\tau\Big) - \frac{p}{H\sqrt{n_0^2-1}}\sin\Big(\frac{n_0^2-1}{H^2}\tau\Big)\\
q(\tau) &= q\cos\omega\tau - \frac{p}{H\omega}\sin\omega\tau
\end{aligned}$$

where, again, we define $\omega = \sqrt{\frac{n_0^2-1}{H^2}}$. Similar analysis for p shows that

$$p(\tau) = p\sin(\omega\tau) + q(H\omega)\cos(\omega\tau)$$

as before. This method of analyzing time evolution lends itself nicely to more complex situations that have no analytical solution. Truncating the Taylor series at a given point gives a sufficiently accurate answer for the motion of light.

Indeed, Reference 2 lays out current directions of research related to understanding the role Lie operators play in Hamiltonian optics. They similarly set up how a discontinuous Galerkin method informed by Lagrangian/Hamiltonian mechanics can be set up in a way that lends itself to optical analysis (see the semi-Lagrangian DG method).

Alternatively, Reference 7 lays out how Lie Transformations can be used to understand the transfer matrix method and derive matrices for more complicated elements on a beam line. Though this reference focuses on applications to accelerator physics, the parallels are clear and further analysis seems appropriate.

Conclusion

Analyzing the GRIN problem in both of these ways elucidates the power of Hamiltonian optics. Similarly to other branches of physics, we have started here by essentially analyzing the optical simple harmonic oscillator. This analysis lays a foundation for future work, and can be used in approximations for more complex systems.

Appendix A

We have the PDE

$$p\frac{\partial S}{\partial q} = -n\frac{\partial n}{\partial q}\frac{\partial S}{\partial p} \tag{11}$$

which does not have well-established boundary conditions. We can still solve the PDE generally and impose some physical understanding to squeeze meaning out of this problem.

We use separation of variables such that $S(q,p) = \xi(q)\chi(p)$ and Equation 11 becomes

$$p\chi(p)\frac{d\xi}{dq} = -n\frac{dn}{dq}\xi(q)\frac{d\chi}{dp}.$$

Separating variables and dividing by S, we find that

$$\frac{1}{-n\frac{dn}{dq}\xi(q)}\frac{d\xi}{dq} = \frac{1}{p\chi(p)}\frac{d\chi}{dp}$$

but the two sides have different dependent variables, so they must separately equal a constant, which we call μ. Then, we have a system of ODES

$$\begin{aligned}\frac{d\xi}{dq} &= -\mu n \frac{dn}{dq}\xi \\ \frac{d\chi}{dp} &= \mu p \chi\end{aligned}$$

which we can solve to find that

$$\begin{aligned}\xi(q) &= C_1 e^{-\mu \frac{n^2}{2}} \\ \chi(p) &= C_2 e^{\mu \frac{p^2}{2}}.\end{aligned}$$

We then combine these to find an expression for S

$$S(q,p) = S_0 \exp\left[\frac{p^2 - n^2}{2}\mu\right], \tag{12}$$

where we assume that initially, S has some known distribution over q and p. Additionally, μ can be any real number without changing the validity of this solution.

References

[1] Wikipedia: Hamiltonian Optics

[2] M. Anthonissen *Hamiltonian optics, Lie algebra and Liouville's equation*, slides

[3] S. T. Thornton, J. B. Marion, *Classical Dynamics of Particles and Systems*, Brooks/Cole Cengage Learning, Boston, 5th ed.

[4] Physics With Elliot, *Before You Start On Quantum Mechanics, Learn This*, video

[5] D. D. Holm, K. B. Wolf, *Lie-Poisson description of Hamiltonian ray optics*, paper

[6] Y. Eidelman, *Introduction to Lie Operators for Accelerator Physics*, lie operators (pp.12-14)

[7] D. Kelliher, *Hamiltonian Dynamics*, presentation

[8] P. Woit, Quantum Field Theory for Mathematicians, lecture notes. Note: consulted minimally for vague understanding of where this work fits within higher-level math and physics